UNIVERSITY COLLEGE LONDON

EXAMINATION FOR INTERNAL STUDENTS

MODULE CODE : **COMP0078**

ASSESSMENT Pattern: **COMP0078A7PE**

MODULE NAME : **Supervised Learning**

LEVEL: : **Postgraduate**

DATE: : **16-May-2023**
TIME : **14:30**
DURATION : **03:15**

Late submission is permitted for Controlled Conditioned exams but late penalties will apply - any submissions that are up to 40 minutes late will be penalised, after which no submissions will be accepted under any circumstances.
You must ensure to allow sufficient time to upload and hand in your work

This paper is suitable for candidates who attended classes for this module in the following academic year(s):

Year
2022-23

Duration	<< Exam Duration>>
Additional time for converting handwritten notes to PDF where applicable	**15 mins**
Upload window	**20**
Total time	**3 Hours 35 mins**

Additional material	N/A
Special instructions	Standard Calculators permitted

TURN OVER

Supervised Learning, COMP0078 (–)

Main Summer Exam Period, 22/23

Suitable for Cohorts: 22/23 (21/22, 20/21, 19/20)

Answer ALL EIGHT questions.

Notation: Let $[m] := \{1, \ldots, m\}$. We also overload notation so that

$$[\texttt{pred}] := \begin{cases} 1 & \texttt{pred} \text{ is true} \\ 0 & \texttt{pred} \text{ is false} \end{cases} .$$

Recommended answer lengths are only for informational purposes, no points will be deducted for shorter or longer answers.

Marks for each part of each question are indicated in square brackets.

Standard calculators are permitted.

1. a. Let $p(x, y)$ denote a probability mass function over (X, Y) where $X = [n]$ and $Y = [k]$ thus $\sum_{x \in X} \sum_{y \in Y} p(x, y) = 1$. For this subquestion let $\boldsymbol{C} \in [0, \infty)^{k \times k}$ be a $k \times k$ matrix of costs. Define $L_{\boldsymbol{C}} : [k] \times [k] \to [0, \infty)$ as,

$$L_{\boldsymbol{C}}(y, \hat{y}) := [y \neq \hat{y}] C_{y,\hat{y}}$$

as the class-sensitive loss function, i.e., if we incorrectly predict $\hat{y}$ and the correct outcome was y we suffer $C_{y,\hat{y}}$ loss. Derive the Bayes estimator.

Now suppose $X = \{1, 2, 3, 4\}$ and $Y = \{1, 2\}$ and

$$\mathbb{P}[X = i] = i/10$$
$$\mathbb{P}[Y = 1|X = i] = 3/10 + i/20$$
$$\mathbb{P}[Y = 2|X = i] = 7/10 - i/20$$

and

$$C = \begin{pmatrix} 0 & 1 \\ 3 & 0 \end{pmatrix}$$

Give the Bayes estimator for this particular problem after performing algebraic simplification.

[5 marks]

b. Given $\mathbf{x}_1, \ldots, \mathbf{x}_m \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^m$ define $A := (y_i y_j \mathbf{x}_i^\top \mathbf{x}_j : i, j \in [m])$ and then let $f : \mathbb{R}^m \to \mathbb{R}$ be defined as $f(\boldsymbol{\alpha}) := \boldsymbol{\alpha}^\top A \boldsymbol{\alpha}$. Is the function f

i. Necessarily convex?

ii. Necessarily concave?

iii. Not necessarily either convex or concave.

Provide a detailed argument justifying your answer.

[5 marks]

2. a. A training set $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_m, y_m)$ is *generic* iff $\mathbf{x}_i = \mathbf{x}_j \Rightarrow y_i = y_j$. Consider the following function $K_a(\mathbf{x}, \mathbf{t}) := \prod_{i=1}^n (1 + (x_i t_i) + (1 - x_i)(1 - t_i))$.

i. Given a generic training set $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_m, y_m) \in \{0,1\}^n \times \{0,1\}$ does there necessarily exist a vector $\boldsymbol{\alpha} \in \mathbb{R}^m$ such that

$$\sum_{j=1}^{m} \left(\sum_{i=1}^{m} \alpha_i K_a(\mathbf{x}_i, \mathbf{x}_j) - y_j \right)^2 = 0\,?$$

[5 marks]

Provide an argument to justify your answer.

ii. Given a generic training set $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_m, y_m) \in \mathbb{R}^n \times \mathbb{R}$ does there necessarily exist a vector $\boldsymbol{\alpha} \in \mathbb{R}^m$ such that

$$\sum_{j=1}^{m} \left(\sum_{i=1}^{m} \alpha_i K_a(\mathbf{x}_i, \mathbf{x}_j) - y_j \right)^2 = 0\,?$$

[5 marks]

Provide an argument to justify your answer.

3. Given the data set,

$$((1,1),+1),((2,2),+1),((2,0),+1),((0,0),-1),((1,0),-1),((0,1),-1)$$

a. Plot the dataset as well as the maximal margin hyperplane. Give the equation of the maximum margin hyperplane. (Note: this may be done by inspection rather than by derivation and proof). Finally, identify the support vector(s).

[5 marks]

b. Consider the following functions from $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$,

i. $f(x,y) = e^{xy}$

ii. $g(x,y) = e^{-xy}$

iii. $k(x,y) = e^{x+y}$

iv. $\ell(x,y) = e^{(x-y)^2}$

v. $m(x,y) = e^{-(x-y)^2}$

Which (if any) of the above functions are "kernels"? Provide arguments supporting your claims. Recall that you may use results stated in the notes without reproving them.

[10 marks]

4. Given a dataset $\mathcal{S} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_m, y_m)\} \subset \mathbb{R}^d \times \mathbb{R}$ and positive scale factors $\mathbf{a} \in (0,\infty)^d$ and $c \in (0,\infty)$ along with offsets $\mathbf{b} \in \mathbb{R}^d$ and $d \in \mathbb{R}$ we define the transforms $f_{\mathbf{a},\mathbf{b}}(\mathbf{x}) := (a_1 x_1 + b_1, a_2 x_2 + b_2, \ldots, a_d x_d + b_d)$ and $g_{c,d}(y) := cy + d$. Consider the regression tree corresponding to the greedy recursive partitioning procedure with respect to square error on the dataset $\mathcal{S}$.

i. Suppose we first transform the dataset to $\{(f_{\mathbf{a},\mathbf{b}}(\mathbf{x}_1), y_1), \ldots, (f_{\mathbf{a},\mathbf{b}}(\mathbf{x}_m), y_m)\}$ then compute the regression tree. Will the computed regression tree change if so how?

ii. Suppose we first transform the dataset to $\{(\mathbf{x}_1, g_{c,d}(y_1)), \ldots, (\mathbf{x}_m, g_{c,d}(y_m))\}$ then compute the regression tree. Will the computed regression tree change if so how?

Give an argument justifying your answers to part i and ii.

[10 marks]

5. In this question we consider *passive* and *aggressive* algorithms in the mistake bound model. An algorithm is passive if it only updates its internal state on a mistaken prediction and it is aggressive if it updates its internal state whether or not there is a mistake. In our online learning notes, the halving algorithm and perceptron algorithm are both passive.

   a. Suppose we apply the halving update on trials also when the prediction is correct. Is it possible to prove the same mistake bound? Give an argument why or why not?

      [5 marks]

   b. Suppose we apply the perceptron update on trials also when the prediction is correct. Is it possible to prove the same mistake bound? Give an argument why or why not?

      [10 marks]

6. a. Recall the following Definition from our discussion on the *agnostic* PAC setting.

Definition (ϵ-representative sample). A training set S is called ϵ-representative if

$$\forall h \in \mathcal{H}, \ \ |L_S(h) - L_{\mathcal{D}}(h)| \leq \epsilon \ .$$

Mathematically, give an analogous definition for the PAC setting. Use (1-4) sentences to explaining the role of these definitions in the proofs of the agnostic pac and the pac sample complexity bound.

[5 marks]

b. Recall the following Lemma.

Lemma. *Assume that a training set S is $\frac{\epsilon}{2}$-representative. Then, any output of* $\mathrm{ERM}_{\mathcal{H}}(S)$*, namely any* $h_S \in \mathrm{argmin}_{h \in \mathcal{H}} L_S(h)$*, satisfies*

$$L_{\mathcal{D}}(h_S) \ \leq \ \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + \epsilon \ .$$

and its proof.

Proof: For every $h \in \mathcal{H}$,

$$L_{\mathcal{D}}(h_S) \overset{1}{\leq} L_S(h_S) + \tfrac{\epsilon}{2} \overset{2}{\leq} L_S(h) + \tfrac{\epsilon}{2} \overset{3}{\leq} L_{\mathcal{D}}(h) + \tfrac{\epsilon}{2} + \tfrac{\epsilon}{2} = L_{\mathcal{D}}(h) + \epsilon \qquad \blacksquare$$

i. Explain why inequality $\overset{1}{\leq}$ is true.

ii. Explain why inequality $\overset{2}{\leq}$ is true.

iii. Explain why inequality $\overset{3}{\leq}$ is true.

[10 marks]

7. Recall the minimum cut method method for semi-supervised learning. Suppose there is an unweighted graph $\mathcal{G}$ with two labeled vertices. A vertex p with the value of "$+1$" and a vertex q with the value "-1". Suppose the value of the minimum cut objective function for this graph is "6". Mathematically that is,

$$6 = \min_{\boldsymbol{u} \in \{-1,1\}^n} \{ \sum_{(i,j) \in E(\mathcal{G})} : |u_i - u_j| : u_p = 1, u_q = -1 \} .$$

a. Draw a partially labeled (unweighted) graph with only two labeled vertices p and q so that if we then calculated the value of the minimum cut objective function it would be "6." Briefly (1-3 recommended sentences) explain why the value of the objective would be "6."

[5 marks]

b. Given the setting of the question. What if anything can we say about the effective resistance between vertex p and q? Explain your reasoning,

[10 marks]

8. a. Consider the following bandit problem with k classes,

Protocol:

Nature selects $\boldsymbol{\ell}_1, \boldsymbol{\ell}_2, \ldots, \boldsymbol{\ell}_T \in \{0,1\}^k$.

For $t = 1$ To T Do

Predict $\hat{y}_t \in [k]$

Receive loss $\ell_{t,\hat{y}_t} \in \{0,1\}$

Design a randomised algorithm with a good upper bound on the expected regret (with respect to the best class),

$$\mathbb{E}\left[\sum_{t=1}^{T} \ell_{t,\hat{y}_t}\right] - \min_{j\in[k]} \sum_{t=1}^{T} \ell_{t,j}$$

give your algorithm design and an argument for your upper bound.

[5 marks]

b. What is a deterministic oblivious adversary? Explain why there are more restrictions on the adversary for EXP3 as opposed to HEDGE.

[5 marks]